

CONTROL / MCU

UCARD

CCARDA1

CCARDA2

CCARDD0

CCARDD1

CCARDD2

CCARDD3

CCARDD4

CCARDVVR

FBCARDA

RCARDRST

GD32G553VET6

The diagram illustrates the pin connections for the GD32G553VET6 microcontroller. It features a central pin header with pins numbered 1 to 100. The connections are as follows:

- UCARD:**
 - V3P3 (pin 8) to pin6
 - WDI (pin 11) to pin11
 - NRST_CARD (pin 14) to pin14
 - AIN9 (pin 16) to pin15
 - AIN10 (pin 18) to pin16
 - AIN8 (pin 17) to pin17
 - AVMID (pin 18) to pin18
 - AIN0 (pin 20) to pin20
 - AIN1 (pin 21) to pin21
 - AIN2 (pin 22) to pin22
 - DGND (pin 23) to pin23
 - V3P3 (pin 24) to pin24
 - AIN11 (pin 25) to pin25
 - AIN12 (pin 26) to pin26
 - AIN3 (pin 27) to pin27
 - EN_B (pin 28) to pin28
 - AIN4 (pin 29) to pin29
 - AIN5 (pin 30) to pin30
 - TSNS0 (pin 32) to pin32
 - TSNS1 (pin 33) to pin33
 - DO6 (pin 34) to pin34
 - AGND (pin 35) to pin35
 - VDDA_CARD (pin 36) to pin36
 - VDDA_CARD (pin 37) to pin37
 - ROLE1 (pin 38) to pin38
 - HMI0 (pin 40) to pin40
 - HMI1 (pin 41) to pin41
 - HMI2 (pin 42) to pin42
 - DIO2 (pin 43) to pin43
 - DIO3 (pin 44) to pin44
 - HMI3 (pin 45) to pin45
 - FLT (pin 47) to pin47
 - DGND (pin 48) to pin48
 - V3P3 (pin 49) to pin49
 - HMI4 (pin 50) to pin50
 - PWM2 (pin 51) to pin51
 - PWM8 (pin 52) to pin52
 - PWM3 (pin 53) to pin53
 - PWM9 (pin 54) to pin54
- CCARDA1:**
 - VDDA_CARD (pin 55) to pin55
 - AIN7 (pin 56) to pin56
 - AIN8 (pin 59) to pin59
 - DO0 (pin 60) to pin60
 - DO1 (pin 62) to pin62
 - DO5 (pin 63) to pin63
 - DGND (pin 64) to pin64
 - V3P3 (pin 65) to pin65
 - PWM5 (pin 66) to pin66
 - PWM11 (pin 67) to pin67
 - PWM4 (pin 68) to pin68
 - PWM10 (pin 69) to pin69
 - PWM0 (pin 70) to pin70
 - PWM6 (pin 71) to pin71
 - PWM1 (pin 72) to pin72
 - PWM7 (pin 73) to pin73
 - DIO4 (pin 74) to pin74
 - DGND (pin 75) to pin75
 - V3P3 (pin 76) to pin76
 - SWDIO_CARD (pin 77) to pin77
 - SWCLK_CARD (pin 78) to pin78
 - DIO5 (pin 81) to pin81
 - DRV_RDY (pin 82) to pin82
 - EN_A (pin 84) to pin84
 - HMI5 (pin 85) to pin85
 - DIO (pin 86) to pin86
 - DIO1 (pin 87) to pin87
 - LINK_TX (pin 88) to pin88
 - LINK_RX (pin 89) to pin89
 - DO2 (pin 90) to pin90
 - ROLE0 (pin 91) to pin91
 - HMI6 (pin 92) to pin92
 - CAN_RX (pin 93) to pin93
 - CAN_TX (pin 94) to pin94
 - DO3 (pin 95) to pin95
 - BOOT0_CARD (pin 96) to pin96
 - DO4 (pin 99) to pin99
 - DGND (pin 100) to pin100
- CCARDA2:**
 - VDDA_CARD (pin 55) to pin55
 - 1uF capacitor (pin 56) to pin56
 - AGND (pin 59) to pin59
- CCARDD0:**
 - VDDA_CARD (pin 55) to pin55
 - 100nF capacitor (pin 56) to pin56
 - DGND (pin 59) to pin59
- CCARDD1:**
 - V3P3 (pin 55) to pin55
 - 100nF capacitor (pin 56) to pin56
 - DGND (pin 59) to pin59
- CCARDD2:**
 - V3P3 (pin 55) to pin55
 - 100nF capacitor (pin 56) to pin56
 - DGND (pin 59) to pin59
- CCARDD3:**
 - V3P3 (pin 55) to pin55
 - 100nF capacitor (pin 56) to pin56
 - DGND (pin 59) to pin59
- CCARDD4:**
 - V3P3 (pin 55) to pin55
 - 100nF capacitor (pin 56) to pin56
 - DGND (pin 59) to pin59
- CCARDVVR:**
 - VDDA_CARD (pin 55) to pin55
 - 100nF capacitor (pin 56) to pin56
 - AGND (pin 59) to pin59
- FBCARDA:**
 - V3P3 (pin 55) to pin55
 - 100nF capacitor (pin 56) to pin56
 - VDDA_CARD (pin 59) to pin59
- RCARDRST:**
 - V3P3 (pin 55) to pin55
 - 10k resistor (pin 56) to pin56
 - NRST_CARD (pin 59) to pin59

CONTROL / SAFETY

The schematic diagram illustrates the control and safety logic, featuring two main integrated circuits (ICs) and several control cards.

U1 (74HC11): A hex inverters IC. Its inputs include EN_A (pin 1), DGND (pin 3), DGND (pin 9), EN_B (pin 2), DGND (pin 4), DGND (pin 10), WDO_CARD (pin 13), DGND (pin 5), and DGND (pin 11). Its output (pin 12) is GATE_EN. It is powered by V3P3 (pin 6) and DGND (pin 7).

U2 (TPS3430): A precision centrer with programmable delay and hysteresis. Its inputs include USUPCARD_CRST (pin 4), USUPCARD_CWD (pin 2), DGND (pin 3), DGND (pin 6), WDI (pin 7), and WDO_CARD (pin 8). Its outputs are CRST (pin 1) and CWD (pin 10). It is powered by V3P3 (pin 5) and DGND (pin 9).

Control Cards:

- CRSTCARD:** Connects USUPCARD_CRST to DGND with a 1nF capacitor.
- CSFCARD:** Connects V3P3 to DGND with a 100nF capacitor.
- CWDPCARD:** Connects USUPCARD_CWD to DGND with a 1nF capacitor.
- RENLCARD:** Connects EN_A to DGND with a 100k resistor.
- RENRCARD:** Connects EN_B to DGND with a 100k resistor.
- RGPDCARD:** Connects GATE_EN to DGND with a 10k resistor.
- RRDYCARD:** Connects V3P3 to DRV_RDY with a 10k resistor.
- RWPUCARD:** Connects V3P3 to WDO_CARD with a 10k resistor.

CONNECTING THE FILTER CAPACITORS

The diagram illustrates three methods for connecting filter capacitors:

- CFLTC:** A 1nF capacitor is connected between the FLT line and the DGND line.
- RAGTC:** A 0R resistor is connected between the AGND line and the DGND line.
- RFLTTC:** A 4.7k resistor is connected between the VSP3 line and the FLT line.

CONTROL / ROLE

The diagram illustrates two control roles, RROLE0 and RROLE1, each represented by a horizontal line with a resistor symbol in the center. The left end of the line is labeled V3P3 and the right end is labeled RROLE0 (for the top) or RROLE1 (for the bottom). Below each resistor is a label '10k'.

The schematic diagram illustrates the control and buck-3V3 circuit for the UBKICARD module. At the top, the **UBKICARD** module is shown as a central component. It has several pins connected to external components:

- Pin 1 (SW):** Connected to **UBKICARD_SW**.
- Pin 2 (BOOT):** Connected to **UBKICARD_BST**.
- Pin 3 (EN):** Connected to **V15**.
- Pin 4 (FB):** Connected to **UBKICARD_FB**.
- Pin 5 (GND):** Connected to **DGND**.

Below the module, the following components are shown in a vertical stack:

- CBKICARD:** A 10 μ F capacitor connected between **V15** and **DGND**.
- CBSTCARD:** A 22 μ F capacitor connected between **V3P3** and **DGND**.
- LBKICARD:** A 100nF capacitor connected between **UBKICARD_BST** and **UBKICARD_SW**.
- IND-10 μ -3A RBKF1CARD:** An inductor connected between **UBKICARD_SW** and **V3P3**.
- 45.3k RBKF2CARD:** A resistor connected between **V3P3** and **UBKICARD_FB**.
- 10k:** A resistor connected between **UBKICARD_FB** and **DGND**.

JCARD	CONN-CARD-88-R
AGND	74
AGND	29
AIN0	13
AIN1	14
AIN2	15
AIN3	16
AIN4	17
AIN5	18
AIN6	19
AIN7	20
AIN8	21
AIN9	22
AIN10	23
AIN11	24
AIN12	25
AVMID	28
CAN_RX	51
CAN_TX	50
DGND	66
DIO	37
DIO	38
DIO	39
DIO	40
DIO	41
DIO	42
DO0	30
DO1	31
DO2	32
DO3	33
DO4	34
DO5	35
DO6	36
AGND	47
AGND_2	48
AIN0	46
AIN1	44
AIN2	43
AIN3	52
AIN4	53
AIN5	54
AIN6	55
AIN7	56
AIN8	57
AIN9	58
AIN10	59
AIN11	60
AIN12	61
AVMID	62
CAN_RX	63
CAN_TX	64
DGND	65
DIO	66
DIO	67
DIO	68
DIO	69
DIO	70
DIO	71
DO0	72
DO1	73
DO2	74
DO3	75
DO4	76
DO5	77
DO6	78
DRV_RDY	47
EN_A	48
EN_B	46
FLT	44
GATE_EN	43
HMI0	52
HMI1	53
HMI2	54
HMI3	55
HMI4	56
HMI5	57
HMI6	58
LINK_RX	49
LINK_TX	48
PWM0	1
PWM1	2
PWM2	3
PWM3	4
PWM4	5
PWM5	6
PWM6	7
PWM7	8
PWM8	9
PWM9	10
PWM10	11
PWM11	12
ROLE0	59
ROLE1	60
TSNS0	26
TSNS1	27
V15	61
VSP3	63